

Supplementary Materials

Benchmarking Recursive-Collapse Warning Claims Under Matched False-Positive Control

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This document contains the supplementary materials referenced in the main manuscript: one figure (S1) and four tables (S2, S3, S4, S5).

Supplementary Figure S1

Supplementary Figure S1. Fast-family segmentation and band sensitivity on the canonical markets benchmark. No fast-family comparator was accepted at 60-, 120-, or 180-minute segmentation under the prespecified equal-false-positive band $[0.03, 0.07]$. Under widened post-processing bands with upper cutoff 0.08, acceptance appeared only for the 60-minute AC1 configuration; the canonical 120-minute and 180-minute segmentations remained non-accepted across all tested bands.

Band	Segmentation	Band reachable	Any fast accepted	Nearest family	Nearest FP	Distance to band
[0.02, 0.08]	60 min	Yes	Yes	ac1_ews	0.0800	0.0000
[0.02, 0.08]	120 min	Yes	No	ac1_ews	0.1316	0.0516
[0.02, 0.08]	180 min	Yes	No	ac1_ews	0.1304	0.0504
[0.03, 0.07]	60 min	Yes	No	ac1_ews	0.0800	0.0100
[0.03, 0.07]	120 min	Yes	No	ac1_ews	0.1316	0.0616
[0.03, 0.07]	180 min	Yes	No	ac1_ews	0.1304	0.0604
[0.04, 0.08]	60 min	Yes	Yes	ac1_ews	0.0800	0.0000
[0.04, 0.08]	120 min	Yes	No	ac1_ews	0.1316	0.0516
[0.04, 0.08]	180 min	Yes	No	ac1_ews	0.1304	0.0504

Source: `results/frozen/comparators/markets_fast_robustness_packet_v1__20260421T175531Z/markets_fast_band_sensitivity_summary_v1.csv`.

Supplementary Table S2

Supplementary Table S2. Effect sizes with bootstrap 95% confidence intervals. Per-witness Cohen’s d, Glass’s d, and rank AUC at each benchmark/horizon (4 benchmarks $\times$ 3 witnesses $\times$ 3 effect measures = 36 cells), with both BCa-adjusted and percentile bootstrap confidence intervals from cluster-aware resampling (10,000 iterations). Markets uses segment-level resampling (n=38 controls + 16 events); recommender (all horizons) uses user-level resampling (n = 40,339 user clusters). Canonical h=50 group is bolded throughout. A sensitivity section flags cells where BCa and percentile CI endpoints differ by ≥ 0.05 (markets p-witness cells only — the bootstrap distribution is right-skewed at low n, and BCa is the more honest read there). Full numerics rendered deterministically by `analysis/17_render_a3_supplementary_table_s2.py` from the per-cell bootstrap output at `results/rendered/effect_sizes/a3_effect_sizes_full.csv`.

Source: `results/rendered/effect_sizes/a3_effect_sizes_full.csv` (commit d8908dd)

Generator: `analysis/17_render_a3_supplementary_table_s2.py`

Per-witness (G, p, δ) effect sizes at each benchmark/horizon with 95% BCa confidence intervals from cluster-aware bootstrap (10,000 iterations). Cohen’s d, Glass’s d, and Rank AUC are reported as complementary magnitude metrics. BCa is the primary CI method; percentile CIs are reported alongside for the cells where they differ meaningfully (sensitivity section below).

Sign convention

Predicted direction by witness (event vs control):

- **G** (amplification): event > control predicted; positive d / AUC > 0.5 confirms
- **p** (concentration): event > control predicted; positive d / AUC > 0.5 confirms
- **δ** (contraction): event < control predicted; negative d / AUC < 0.5 confirms

Bootstrap unit grain

- **Markets:** segment-level resampling (n = 38 controls + 16 events; ~30 rows per segment)
- **Recsys** (all horizons): user-level resampling (n = 40,339 user clusters; 10 rows per cluster)

Effect sizes by benchmark

Markets (Volmageddon + COVID-MWCB)

Witness	Cohen’s d [BCa 95% CI]	Glass’s d [BCa 95% CI]	Rank AUC [BCa 95% CI]
G	-0.029 [-0.169, +0.147]	-0.028 [-0.155, +0.160]	0.507 [0.459, 0.555]
p	+0.097 [-0.488, +0.941]	+0.119 [-0.412, +2.094]	0.482 [0.398, 0.614]
δ	+0.029 [-0.568, +0.617]	+0.029 [-0.563, +0.617]	0.516 [0.353, 0.680]

Recsys h=40 (off-pre-reg)

Witness	Cohen's d [BCa 95% CI]	Glass's d [BCa 95% CI]	Rank AUC [BCa 95% CI]
G	-0.214 [-0.235, -0.192]	-0.213 [-0.233, -0.192]	0.441 [0.436, 0.447]
p	+0.180 [+0.155, +0.205]	+0.172 [+0.147, +0.197]	0.555 [0.549, 0.562]
δ	-0.268 [-0.299, -0.237]	-0.219 [-0.245, -0.190]	0.415 [0.407, 0.423]

Recsys h=50 (canonical, pre-registered)

Witness	Cohen's d [BCa 95% CI]	Glass's d [BCa 95% CI]	Rank AUC [BCa 95% CI]
G	+0.100 [+0.076, +0.124]	+0.112 [+0.084, +0.139]	0.519 [0.513, 0.525]
p	+0.080 [+0.049, +0.111]	+0.073 [+0.043, +0.102]	0.531 [0.523, 0.539]
δ	-0.168 [-0.208, -0.128]	-0.125 [-0.155, -0.095]	0.457 [0.447, 0.467]

Recsys h=60 (off-pre-reg)

Witness	Cohen's d [BCa 95% CI]	Glass's d [BCa 95% CI]	Rank AUC [BCa 95% CI]
G	+0.333 [+0.310, +0.354]	+0.453 [+0.411, +0.495]	0.575 [0.570, 0.580]
p	+0.007 [-0.028, +0.043]	+0.006 [-0.026, +0.038]	0.513 [0.504, 0.522]
δ	-0.029 [-0.072, +0.016]	-0.021 [-0.053, +0.011]	0.508 [0.497, 0.520]

BCa-vs-percentile sensitivity

BCa correction matters where the bootstrap distribution is skewed (small n; right-skewed tails on small-n p-witness). Cells where $|\text{BCa endpoint} - \text{percentile endpoint}| \geq 0.05$ on either side:

Benchmark	Witness	Measure	Percentile CI	BCa CI	max Δ
Markets	p	glasss_d	[-0.492, +1.356]	[-0.412, +2.094]	0.74
Markets	p	cohens_d	[-0.589, +0.695]	[-0.488, +0.941]	0.25

Notes

- **Recsys BCa correction is empirically negligible.** Across all 27 recsys cells at $n = 40,339$, $|\text{BCa} - \text{percentile}| < 0.01$ on every endpoint. Consistent with BCa's $O(1/\sqrt{n})$ second-order correction theory: at this n the correction is below the bootstrap Monte Carlo noise floor. Percentile CIs would suffice for recsys; BCa is reported for completeness and procedural consistency with markets.

- **Markets BCa correction is substantive for the p witness.** The Glass’s d markets p cell upper bound shifts from percentile +1.357 to BCa +2.094 (54% right-shift). Bootstrap distribution is right-skewed at n=54; percentile CIs systematically understate the right tail. BCa is the honest read here.
- **Canonical h=50 is the only configuration where all three witnesses align in the predicted direction with CIs clear of 0** (bolded above). Adjacent recsys horizons degrade asymmetrically: h=40 G flips sign while p and δ hold; h=60 G strengthens while p and δ collapse to null. The pre-registered horizon delivers the cleanest signature — evidence that pre-registration was load-bearing for the bridge claim.
- **Markets row-level effects are uniformly null** (all three witnesses, all three measures, all CIs span 0 or 0.5). The directional bridge claim for markets operates at unit-level aggregation grain (mean per unit, then compare unit means across the n=38+16 segments); the row-level effect-size table above is honest data at the finer grain. See main text Cross-domain evidence section for the unit-level framing.

See also: Figure 6 (effect-size forest plot, results/figures/a3_effect_size_forest.{png,pdf}); Cross-domain evidence section of the main manuscript.

Supplementary Table S3

Supplementary Table S3. Alert-count exact-matching sensitivity check. Re-expression of the envelope-boundary matched-FP comparison (main-text Table 2) in integer alarm counts. For each panel at the canonical $k = 3$, Loopzero is reported at its envelope-boundary alarm counts (`n_control_alarmed`, `n_event_alarmed`); each comparator family’s frozen pre-registered calibration grid is searched for the operating point with the nearest `n_control_alarmed` value, with status classified as `exact_match` (comparator has an operating point at Loopzero’s exact alarm count), `bounded_gap` (Loopzero’s alarm count falls between two adjacent comparator operating points, with `gap` quantifying the calibration coverage gap), `no_overlap_above` (all comparator operating points have higher alarm counts), or `insufficient_data`. On the markets benchmark, `ac1_ews` exhibits `exact_match` at Loopzero’s envelope-boundary anchor: both methods alarm on exactly 9 of 38 controls and detect exactly 3 of 16 events. On the recommender benchmark, `variance_ews` exhibits the widest bounded gaps across Loopzero’s envelope boundary: 3559 alarms at $h = 40$ (Loopzero at 97 of 6562 controls) and 3345 alarms at $h = 60$ (Loopzero at 2 of 3996 controls), in each case with the upper-bound comparator operating point detecting all panel event units ($\text{TPR} = 1.0$). At $h = 50$, no comparator family produced a non-trivial breakpoint at Loopzero’s envelope-boundary FP of 0.002313; accordingly, $h = 50$ is reported for transparency but excluded from the cross-horizon comparator-envelope comparison. Full numerics rendered by `analysis/21_compute_a2_alert_count_exact.py` from frozen calibration data at `results/rendered/a4_roc_lowfp/a4_roc_data.parquet` and operating-point data at `results/rendered/bridge/a1_loopzero_operating_points{,_h40_h60}.csv`. Columns renamed in this rendering for compactness: `lower` and `upper` abbreviate `comp_lower` and `comp_upper`; `gap` abbreviates `gap_width_n_control`; `n_bp` abbreviates `n_breakpoints`.

Companion to `a2_threshold_path_envelope_boundary.{csv,md}`. Re-frames the envelope-boundary comparison in integer alarm counts to surface (a) discrete-FP-space coincidences as exact

alarm-count matches, and (b) linear-interpolation gaps as quantitative gap widths in alarm-count space.

Comparator alarm counts derived from `round(fp_comp * panel.n_control_units)` and `round(tpr_comp * panel.n_event_units)`. Status legend:

- `loopzero_envelope_boundary` — Loopzero’s anchor at panel max FP, $k=3$
- `exact_match` — comparator has a breakpoint at exactly Loopzero’s alarm count
- `bounded_gap` — Loopzero’s alarm count falls strictly between two adjacent comparator breakpoints; `gap` records the gap size
- `no_overlap_above` — all comparator breakpoints have higher alarm counts
- `no_overlap_below` — all comparator breakpoints have lower alarm counts
- `insufficient_data` — comparator has <1 useful breakpoint

Panel: markets

Panel `n_control_units=38`, `n_event_units=16`. Loopzero at envelope boundary ($q=50$, $k=3$): **9 false alarms / 38 controls** (FP=0.236842), **3 true detections / 16 events** (TPR=0.1875).

family	status	lower (ctrl, evt)	upper (ctrl, evt)	gap	n_bp
ac1_ews	exact_match	(9, 3)	(9, 3)	0	25
cusum	no_overlap_above	—	(14, 5)	—	15
page_hinkley	no_overlap_above	—	(26, 14)	—	6
variance_ews	no_overlap_above	—	(17, 5)	—	20

Panel: recsys_h40

Panel `n_control_units=6562`, `n_event_units=33777`. Loopzero at envelope boundary ($q=50$, $k=3$): **97 false alarms / 6562 controls** (FP=0.014782), **3113 true detections / 33777 events** (TPR=0.0922).

family	status	lower (ctrl, evt)	upper (ctrl, evt)	gap	n_bp
ac1	no_overlap_above	—	(4206, 33777)	—	3
cusum	no_overlap_above	—	(4010, 33777)	—	4
matrix_profile	no_overlap_above	—	(155, 3938)	—	2
page_hinkley	no_overlap_above	—	(4206, 33777)	—	4
permutation_entropy	no_overlap_above	—	(6562, 33777)	—	1
variance_ews	bounded_gap	(0, 0)	(3559, 33777)	3559	6

Panel: recsys_h50

Panel `n_control_units=4755`, `n_event_units=35584`. Loopzero at envelope boundary ($q=50$, $k=3$): **11 false alarms / 4755 controls** (FP=0.002313), **2681 true detections / 35584 events** (TPR=0.0753).

family	status	lower (ctrl, evt)	upper (ctrl, evt)	gap	n_bp
<i>no comparator breakpoint at this boundary FP</i>	insufficient_data	—	—	—	0

Panel: recsys_h60

Panel n_control_units=3996, n_event_units=36343. Loopzero at envelope boundary (q=50, k=3): **2 false alarms / 3996 controls** (FP=0.000501), **2004 true detections / 36343 events** (TPR=0.0551).

family	status	lower (ctrl, evt)	upper (ctrl, evt)	gap	n_bp
ac1	no_overlap_above	—	(3464, 36343)	—	3
cusum	no_overlap_above	—	(3430, 36343)	—	4
matrix_profile	no_overlap_above	—	(35, 3938)	—	2
page_hinkley	no_overlap_above	—	(3464, 36343)	—	4
permutation_entropy	no_overlap_above	—	(3996, 36343)	—	1
variance_ews	bounded_gap	(0, 0)	(3345, 36343)	3345	6

Supplementary Table S4

Supplementary Table S4. Cluster-robust sensitivity at scenario grain (markets benchmark). Per-witness effect sizes (Cohen’s d, Glass’s d, rank AUC) on the markets benchmark at the A3 segment grain (n = 54 segments) and the B3 scenario grain (n = 7 clusters: 2 event scenarios — `volmageddon_2018_xiv` with 8 segments and `covid_mwcb_2020_03_18` with 8 segments; 5 control scenarios — `covid_noncollapse_2020_03_11` with 8 segments, `volmageddon_control_2018_01_29` with 8 segments, `volmageddon_control_2018_02_08` with 8 segments, `covid_noncollapse_2020_03_13` with 7 segments, `volmageddon_control_2018_01_25` with 7 segments). Bootstrap iterations: 10,000 per cell; CI level: 0.95; RNG seed for scenario grain: 43 (distinct from A3 segment grain seed: 42). For p|Glass’s d at scenario grain, 2 of 10,000 iterations produced degenerate samples (control SD ≈ 0 from same-scenario cluster resamples); finite-filter applied to remaining 9,998 iterations. All other 8 cells: n_degenerate = 0. Scenario-grain CIs are reported as sensitivity, not primary inference; the segment-grain baseline (Supplementary Table S2) remains the primary report.

Cluster composition

Event-side scenario clusters (derived from `unit_id` parsing; n=2):

Cluster	Segments
<code>volmageddon_2018_xiv</code>	8

Cluster	Segments
covid_mwcb_2020_03_18	8

Control-side scenario clusters (n=5):

Cluster	Segments
covid_noncollapse_2020_03_11	8
volmageddon_control_2018_01_29	8
volmageddon_control_2018_02_08	8
covid_noncollapse_2020_03_13	7
volmageddon_control_2018_01_25	7

Total cluster units: 7 (2 event + 5 control), down from 54 segment units in A3.

Methodological framing

Cluster-aware bootstrap (10,000 iterations) at scenario grain. Reported as a **dependence-sensitivity diagnostic** at $n = 2$ event clusters; at this sample size the bootstrap intervals are not reliable conservative inferential intervals. Wild cluster bootstrap (Cameron, Gelbach & Miller 2008) — the methodologically appropriate response to the small-cluster regime — is deferred to follow-up work.

Effect sizes at scenario grain (B3)

Witness	Measure	Point	Percentile 95% CI	BCa 95% CI
G	cohens_d	-0.0293	[-0.0916, +0.0489]	[-0.0932, +0.0473]
G	glasss_d	-0.0285	[-0.0934, +0.0472]	[-0.0938, +0.0472]
G	rank_auc	+0.5073	[+0.4569, +0.5581]	[+0.4520, +0.5530]
p	cohens_d	+0.0966	[-0.3309, +0.4082]	[-0.3285, +0.4313]
p	glasss_d	+0.1188	[-0.2952, +0.7102]	[-0.2950, +0.7102]
p	rank_auc	+0.4823	[+0.4376, +0.5310]	[+0.4361, +0.5305]
delta	cohens_d	+0.0286	[-0.1534, +0.1993]	[-0.1534, +0.1993]
delta	glasss_d	+0.0287	[-0.1610, +0.1928]	[-0.1610, +0.1928]
delta	rank_auc	+0.5162	[+0.4495, +0.5773]	[+0.4495, +0.5757]

Side-by-side: A3 segment grain vs B3 scenario grain (BCa 95% CI)

Witness	Measure	A3 segment grain (n=54)	B3 scenario grain (n=7)	CI width ratio
G	cohens_d	[-0.1686, +0.1469]	[-0.0932, +0.0473]	0.45×
G	glasss_d	[-0.1552, +0.1598]	[-0.0938, +0.0472]	0.45×
G	rank_auc	[+0.4592, +0.5552]	[+0.4520, +0.5530]	1.05×
p	cohens_d	[-0.4884, +0.9409]	[-0.3285, +0.4313]	0.53×

Witness	Measure	A3 segment grain (n=54)	B3 scenario grain (n=7)	CI width ratio
p	glasss_d	[-0.4123, +2.0944]	[-0.2950, +0.7102]	0.40×
p	rank_auc	[+0.3979, +0.6141]	[+0.4361, +0.5305]	0.44×
delta	cohens_d	[-0.5682, +0.6167]	[-0.1534, +0.1993]	0.30×
delta	glasss_d	[-0.5632, +0.6174]	[-0.1610, +0.1928]	0.30×
delta	rank_auc	[+0.3527, +0.6799]	[+0.4495, +0.5757]	0.39×

Supplementary Table S5. Reproducibility anchor hashes

Freeze-manifest SHA-256 hashes for the empirical components anchoring all results in this paper. Each frozen directory additionally contains a LOCK_NOTE.md documenting that freeze’s scientific and editorial scope.

Component	Freeze path	SHA-256
Recsys benchmark freeze state (canonical)	results/frozen/movielens 25m_recursive_frontier_public_v1__benchmark_freeze_state.json	050aa725e6608f659b3f7eb1 03731bb7fbabaa23aeefdc4b 11043999b7fed4cf
Recsys contract freeze	results/frozen/movielens 25m_recursive_frontier_public_v1__contract_freeze_state.json	f361d937cc07cbb946753ac9 e5e2d1d386c8706a6bb43841 229ec846fd69ecd6
Recsys manuscript freeze state	results/frozen/movielens 25m_recursive_frontier_public_v1__manuscript_freeze_state.json	bc8542cc0b76cdfad9726a40 85c760d38ed7ace24772f0e3 c6103fba962891b8
Markets comparator merged state (v2)	results/frozen/comparators/markets_comparator_merged_state_v2__20260421T174702Z/freeze_manifest.json	784895418953b1646bd40eb7 b5266e39901326b1335b2ec5 f2390b70af4642c7
Markets fast robustness packet (v1)	results/frozen/comparators/markets_fast_robustness_packet_v1__20260421T175531Z/freeze_manifest.json	f8f8c5c58d10796d4b715452 82a8b352c47dc9590fc268fc b96cfb84c181adec
Witness-direction bridge state (v3, canonical)	results/frozen/bridge/witness_direction_bridge_state_v3__20260422T130759Z/freeze_manifest.json	bd47e75851fcaea1e6d843f9 b74156ec036ddb487ca76e81 5d7b6fb0cf258f94
Manuscript two-domain state (v1)	results/frozen/manuscript/two_domain_paper_state_v1__20260422T134303Z/freeze_manifest.json	1763b04cb128a274dd12313a 8cccd8b2a04e2845d176d3444 d6214e0aaf2e1d5

Component	Freeze path	SHA-256
Recsys horizon sensitivity manifest (covers h=40, h=50, h=60)	results/manifests/movielens25m_recursive_frontend_public_v1__horizon_sensitivity_summary.json	b5d7ef3b5ca10f9e446a9792bc7d677a323913c66c087e753c6de944a8b7f215
MovieLens-25M source archive (upstream)	https://files.grouplens.org/datasets/movielens/ml-25m.zip (GroupLens; not redistributed)	8b21cfb7eb1706b4ec0aac894368d90acf26ebdfb6aced3ebd4ad5bd1eb9c6aa

Engine: movielens_recursive_replay_engine v1.0.0 (item_item_collaborative_filtering, hash 56c1cff225d60c09). Recsys contract: SHA-256 2e256b255a7f074c1516d70315ebb216241a4a7e8aba2db88b194417705fd71d. All hashes are SHA-256 in lowercase hexadecimal.

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